

27 Light from a laser is incident normally on a double slit. A wave of light emerges from each slit.

Bright and dark interference fringes are observed on a screen that is placed parallel to the double slit.

Which statement explains the formation of a dark fringe?

- A** A dark fringe is formed where none of the light reaches the screen because it was blocked at the double slit.
- B** A dark fringe is formed where the waves of light from the two slits do **not** overlap and so **cannot** superpose.
- C** A dark fringe is formed where a trough from one wave of light meets a trough from the other wave of light.
- D** A dark fringe is formed where the two waves of light meet and have a phase difference of 180° .